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Chapter 9 — Human Digestive System

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter has one main textbook topic, 9.1 Anatomy and Physiology of Digestive System. It is converted into board-style long questions covering each organ and process, followed by conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 9 carries 1 MCQ, 3 short questions in Q.3, and part (b) of a long question in Q.6. Shorts are 2 marks; longs are 8 marks.

TOPIC 9.1 — ANATOMY AND PHYSIOLOGY OF DIGESTIVE SYSTEM

Textbook topic 9.1 · printed pages 202–212

Q.1 Describe the oral cavity and the mechanical and chemical digestion of food.

Board long question · 8 marks

The human digestive system consists of the **gastrointestinal (GI) tract** and **accessory digestive organs**. The GI tract extends from mouth to anus and includes oral cavity, pharynx, oesophagus, stomach, small intestine and large intestine. Accessory organs include salivary glands, liver, gallbladder and pancreas.

Oral cavity

- **Selection of food:** The tongue, taste buds, smell and sight help select or reject food.
- **Mechanical digestion:** Teeth break food into smaller pieces by **mastication**, increasing surface area for enzymatic action.
- **Chemical digestion:** Salivary glands pour saliva into the oral cavity. Saliva contains water and mucus for lubrication, **bicarbonate ions** for buffering, **thiocyanate ions** that kill microorganisms, and **salivary amylase**, which partially digests starch and glycogen to maltose.

The three pairs of salivary glands are **parotid** (in front of ears), **submaxillary** (behind jaws) and **sublingual** (below tongue). After chewing and mixing with saliva, food becomes a small moist mass called a **bolus**.

Q.2 Explain swallowing and the motility of the alimentary canal.

Board long question · 8 marks

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Swallowing

The tongue moves upwards and backwards, forcing the bolus to the back of the oral cavity. The soft palate raises against the back wall of the pharynx, closing the nasal passage. The larynx moves upward and the **epiglottis** closes the trachea, so the bolus passes into the oesophagus. The beginning of swallowing is voluntary, but once food reaches the pharynx it becomes automatic.

Motility

- **Peristalsis:** Rhythmic waves of smooth-muscle contraction that squeeze food along the alimentary canal.
- **Segmentation:** Repeated contraction and relaxation of smooth-muscle rings in the intestines, creating back-and-forth movement that mixes food with digestive secretions and increases absorption efficiency.
- **Antiperistalsis:** Reversal of peristalsis, pushing food back into the stomach and oral cavity, leading to vomiting.

The oesophagus is about **25 cm** long. The **lower oesophageal (cardiac) sphincter** controls entry into the stomach and prevents backflow.

Q.3 Describe the structure of the stomach and the regulation of gastric juice secretion.

Board long question · 8 marks

The stomach is an elastic muscular **U-shaped** bag between the oesophagus and duodenum. It has three portions: **cardiac**, **fundus** and **pyloric**. Its wall has four layers: mucosa, submucosa, muscularis externa and serosa. Muscular contractions churn food and mix it with gastric secretions.

Gastric glands

- **Mucous cells:** Secrete mucus that protects the stomach lining from HCl and enzymes.
- **Parietal (oxyntic) cells:** Secrete **HCl**, adjusting pH to about **2–3**, softening food, activating pepsinogen and killing microorganisms.
- **Chief cells:** Secrete **pepsinogen**, which is activated to **pepsin** by HCl. Pepsin breaks proteins into polypeptides and peptides.

Regulation

The smell, sight or thought of food stimulates the medulla to send nervous signals for gastric secretion. When food enters, stomach distension and low pH increase secretion. Proteins stimulate endocrine cells to release **gastrin**, which stimulates gastric glands. When food moves to the small intestine, **somatostatin** stops the release of HCl. The semi-liquid acidic mass formed is called **chyme**, released through the pyloric sphincter.

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Q.4 Describe digestion in the small intestine. What are the roles of pancreatic juice and bile?

Board long question · 8 marks

The small intestine is about **6 m** long and consists of **duodenum**, **jejunum** and **ileum**. It is responsible for final digestion and absorption.

Duodenum

The duodenum receives **pancreatic juice** and **bile**. Pancreatic juice is slightly alkaline (pH 8) due to bicarbonate and neutralizes acidic chyme. It contains:

- **Pancreatic amylase:** Digests polysaccharides to maltose and glucose.
- **Trypsinogen:** Activated by enterokinase to trypsin, which digests proteins.
- **Chymotrypsin and carboxypeptidase:** Digest proteins into smaller peptides and amino acids.
- **Pancreatic lipase:** Digests lipids to glycerol and fatty acids.
- **Pancreatic nucleases:** Digest DNA and RNA into nucleotides.

Bile, produced by the liver and stored in the gallbladder, contains bile salts that **emulsify fats** into small droplets, increasing surface area for lipase action. Bile does not contain digestive enzymes and does not neutralize acid; pancreatic bicarbonate neutralizes the chyme.

Jejunum and ileum

Intestinal juice contains aminopeptidase, erepsin, lipase, maltase, sucrase and lactase, which complete digestion. The chyme becomes an alkaline emulsion called **chyle**.

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Q.5 Explain the absorption of digested food in the small intestine.

Board long question · 8 marks

Absorption occurs mainly in the jejunum and ileum. The inner wall has circular folds, **villi** and **microvilli**, which greatly increase surface area. Each villus has blood capillaries and a lymphatic vessel called a **lacteal**.

- **Simple sugars and amino acids:** Absorbed by diffusion or active transport into epithelial cells, then into blood capillaries. The **hepatic portal vein** carries them to the liver, where excess glucose is stored as glycogen and amino acids as proteins.
- **Fatty acids and glycerol:** Absorbed by passive transport into epithelial cells, where they are reassembled into triglycerides and coated with proteins to form **chylomicrons**. Chylomicrons enter lacteals, move through the thoracic lymphatic duct and enter the bloodstream.

Q.6 Describe the large intestine and the control of egestion.

Board long question · 8 marks

The large intestine is about the last metre of the intestinal tract. It absorbs **water, salts and vitamin K**. It has no villi and consists of caecum, colon and rectum.

- **Caecum:** A blind pouch with the vestigial **vermiform appendix**.
- **Colon:** Ascending, transverse and descending limbs; absorbs water, making faeces more solid. Faeces contain bacteria, indigestible fibre, sloughed cells, bile pigments and water.
- **Rectum:** Stores faeces temporarily. The anus has internal (smooth muscle) and external (striated muscle) sphincters.

In infants, egestion is an **involuntary spinal reflex**. In adults, maturation of the cerebral cortex allows **voluntary control** of the external anal sphincter. Bacteria such as E. coli in the colon produce **vitamin K**, which is necessary for blood coagulation and is absorbed into the blood.

Q.7 Describe the accessory digestive organs and the hormonal control of their secretions.

Board long question · 8 marks

Liver and gallbladder

The liver produces **bile**, which emulsifies fats. It also processes absorbed nutrients, detoxifies harmful substances, synthesizes proteins and stores glycogen, vitamins and minerals. The gallbladder stores and concentrates bile and releases it into the duodenum.

Pancreas

The pancreas has an **exocrine** portion that secretes pancreatic juice containing enzymes for all food groups, and an **endocrine** portion that secretes **insulin** and **glucagon** into the blood.

Hormonal control

When acidic chyme enters the duodenum, it stimulates release of **secretin**. Partially digested proteins and fats stimulate release of **cholecystokinin (CCK)**. Both hormones stimulate the pancreas to release pancreatic juice and the gallbladder to release bile.

Storage and metabolic role of liver

The liver stores glucose as glycogen, vitamins A, D, B12 and K, minerals such as iron and copper, and fats. It converts excess glucose to glycogen, breaks down fats, forms cholesterol and lipoproteins, deaminates amino acids and converts toxic ammonia to **urea**. It also detoxifies drugs, alcohol and hormones.

Important Conceptual Short Questions — Topic 9.1

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S1. How does mastication increase the efficiency of digestion?

It breaks food into smaller pieces, increasing surface area for enzymatic action.

S2. Why does saliva contain bicarbonate and thiocyanate ions?

Bicarbonate buffers chemicals in the oral cavity, while thiocyanate ions kill microorganisms.

S3. How does the epiglottis prevent food from entering the trachea?

During swallowing, the larynx moves upward and the epiglottis closes the tracheal opening, directing the bolus into the oesophagus.

S4. Differentiate between peristalsis and segmentation.

Peristalsis propels food forward by waves of contraction. Segmentation mixes food with digestive secretions by back-and-forth movement.

S5. Why is the stomach lining not digested by its own enzymes?

Mucous cells secrete mucus that protects the mucosa from HCl and pepsin.

S6. How does HCl help protein digestion in the stomach?

It activates pepsinogen to pepsin and provides the acidic pH needed for pepsin activity.

S7. Why is pepsinogen secreted instead of active pepsin?

Active pepsin could digest the proteins of the stomach wall, so it is secreted in inactive form and activated only in the stomach lumen.

S8. How does gastrin regulate gastric secretion?

It is released in response to proteins in food and stimulates gastric glands to secrete more gastric juice.

S9. Why does somatostatin stop acid release when food enters the small intestine?

It prevents excessive acidity and helps coordinate digestion as chyme moves from stomach to intestine.

S10. How does bile assist fat digestion if it contains no enzymes?

Bile salts emulsify fats into small droplets, increasing surface area for pancreatic lipase.

S11. Why does pancreatic juice contain bicarbonate?

It neutralizes the acidic chyme entering the duodenum, providing a suitable pH for intestinal and pancreatic enzymes.

S12. How do villi and microvilli increase absorption?

They greatly increase the internal surface area of the small intestine, allowing more efficient uptake of digested food.

S13. Why do fatty acids enter lacteals instead of blood capillaries directly?

They are reassembled into triglycerides and packaged as chylomicrons, which are too large to enter blood capillaries directly and enter lymphatic lacteals instead.

S14. How does the hepatic portal vein help regulate nutrient levels?

It carries absorbed sugars and amino acids to the liver, where excess nutrients are stored or processed before entering general circulation.

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S15. Why is vitamin K absorption important in the large intestine?

Vitamin K is necessary for blood coagulation, and bacteria in the colon produce it for absorption into the blood.

S16. How does egestion change from infancy to adulthood?

In infants it is an involuntary spinal reflex. In adults the cerebral cortex matures, allowing voluntary control of the external anal sphincter.

S17. Why is the pancreas considered both an exocrine and endocrine gland?

Its exocrine portion secretes pancreatic juice into a duct, while its endocrine portion secretes insulin and glucagon into the blood.

S18. How do secretin and CCK coordinate digestion in the duodenum?

Secretin responds to acidity and CCK to proteins/fats; both stimulate pancreatic juice and bile release.

S19. How does the liver protect the body from ammonia toxicity?

It converts harmful ammonia produced from amino acid breakdown into urea, which is excreted in urine.

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